

PLANT PATHOLOGY

Evolution of the wheat blast fungus through functional losses in a host specificity determinant

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Wheat blast first emerged in Brazil in the mid-1980s and has recently caused heavy crop losses in Asia. Here we show how this devastating pathogen evolved in Brazil. Genetic analysis of host species determinants in the blast fungus resulted in the cloning of avirulence genes *PWT3* and *PWT4*, whose gene products elicit defense in wheat cultivars containing the corresponding resistance genes *Rwt3* and *Rwt4*. Studies on avirulence and resistance gene distributions, together with historical data on wheat cultivation in Brazil, suggest that wheat blast emerged due to widespread deployment of *rwt3* wheat (susceptible to *Lolium* isolates), followed by the loss of function of *PWT3*. This implies that the *rwt3* wheat served as a springboard for the host jump to common wheat.

Host jumps of plant pathogens may cause outbreaks of new crop diseases. A recent, likely example is wheat blast caused by *Pyricularia oryzae* (*Magnaporthe oryzae*). *P. oryzae* is composed of host-specific subgroups such as the *Oryza*, *Eleusine*, *Avena*, and *Lolium* pathotypes that cause disease in rice, finger millet, oat, and perennial ryegrass, respectively (1–3). Wheat blast was first reported in 1985 in Paraná state in the southern region of Brazil; it then spread to wheat-growing areas in neighboring states and countries in South America (4). The causal agent (the wheat blast pathogen) was identified as *P. oryzae* (4) but was specifically pathogenic on wheat and its wild relatives (*Triticum* spp. and *Aegilops* spp.) (3, 5); therefore, the pathogen is considered to be a previously unrecognized host-specific subgroup (*Triticum* pathotype). In 2011, wheat blast was found in North America, at a University of Kentucky research plot (6). Though less virulent than South American isolates, the strain isolated in Kentucky was pathogenic on wheat and was inferred to have evolved from annual ryegrass pathogen in North America through host jump (6). In 2016, wheat blast suddenly appeared in Bangladesh and

caused a substantial loss of wheat production (7, 8). Phylogenomic analyses revealed that this outbreak was most likely caused by a wheat-infecting strain from South America (7, 8). Currently, this devastating disease has taken a major step toward becoming pandemic and poses a serious threat to global wheat production. Here we show that the wheat blast pathogen evolved through functional losses in a host specificity determinant.

Triticum isolates are most closely related to *Avena* and *Lolium* isolates (Fig. 1A). Takabayashi

et al. (9) identified two genes, *PWT3* and *PWT4*, in *Avena* isolate Br58, which conditioned its avirulence on wheat. They also identified the resistance gene in wheat that recognizes *PWT4* and designated it as *Rmg1* (synonymous with *Rwt4*). Similarly, Vy *et al.* (10) identified a gene (tentatively named *A1*) playing the primary role in the avirulence of *Lolium* isolate TP2 on wheat, as well as its corresponding resistance gene *Rmg6*. Allelism tests revealed that *PWT3* is located at the same locus as *A1* (fig. S1) and is recognized by *Rmg6* (fig. S2). These data suggest that the same gene pair, *PWT3* and *Rmg6* (syn. *Rwt3*), is involved in the incompatibility of both *Lolium* and *Avena* isolates on wheat (fig. S3).

We isolated *PWT3* from *Avena* isolate Br58 through map-based cloning (fig. S4). The *PWT3* nucleotide sequence from Br58 (later designated as A type) was shared by all 12 *Lolium* isolates analyzed, including TP2 (table S1), supporting the status of *PWT3* as a host species specificity gene. We also isolated *PWT4* from Br58 by using bulked segregant analysis coupled with whole-genome sequencing (fig. S4). The predicted *PWT3* and *PWT4* proteins contained putative signal peptides (fig. S4) but lacked similarity to known proteins or protein domains.

PWT3 and *PWT4* were identified through seedling infection assays, but wheat blast is mainly a spike disease in Brazilian wheat fields. To estimate the roles of these genes during field infections, spikes of wheat cultivars Norin 4 (N4) (*Rwt3/Rwt4*), Chinese Spring (CS) (*Rwt3/rwt4*), Transfed (Tfed) (*rwt3/Rwt4*), and Hope (*rwt3/rwt4*) were inoculated at early anthesis with *Triticum* isolate Br48 and transformants carrying *PWT3* (Br48+3) and *PWT4* (Br48+4). Br48 was virulent on spikes of all four cultivars, whereas Br48+3 and Br48+4

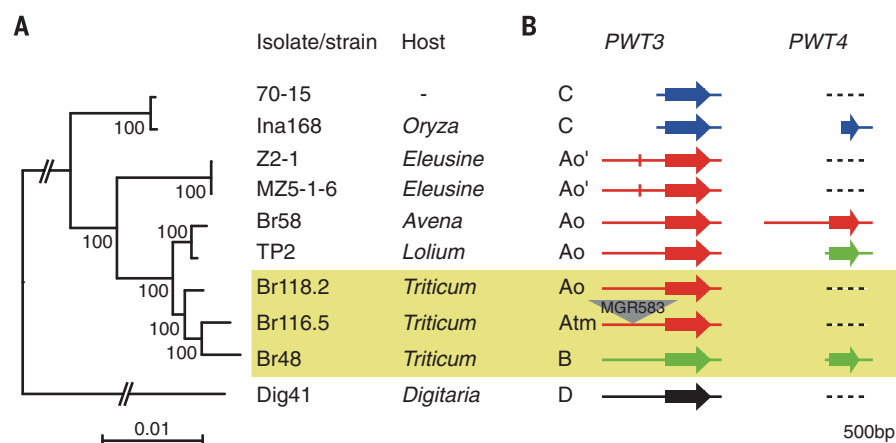


Fig. 1. Distribution of *PWT3* and *PWT4* in *Pyricularia* spp. (A) Maximum likelihood tree of *P. oryzae* isolates constructed from SNPs in whole-genome sequences. *P. grisea* (Dig41) was used as an outgroup. The numbers on the branches indicate bootstrap probability. The bar below the tree indicates genetic distance per site. **(B)** Schematic representation of *PWT3* and *PWT4* types among *Triticum* isolates (highlighted in the shaded region) from the 1990s (Kobe University collection) and other pathotypes. The arrows and horizontal bars represent ORFs and flanking regions, respectively. For *PWT3*, Ao and B are avirulent and virulent types, respectively. The A' and Atm types are identical to the Ao type, except for a one-base substitution and an insertion of retrotransposon MGR583 (gray triangle) in the upstream region, respectively. For *PWT4*, red, green, and blue arrows represent the avirulent type, the virulent type, and a truncated virulent type, respectively. Dotted lines indicate the absence of homologs. bp, base pairs.

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showed specific avirulence on the *Rwt3* and *Rwt4* carriers, respectively (Fig. 2A). Thus, *PWT3-Rwt3* and *PWT4-Rwt4* interactions play critical roles in spike infection.

We performed gene-disruption experiments to determine whether *PWT3-Rwt3* and *PWT4-Rwt4* interactions are the only barriers preventing the nonwheat isolates from infecting *Rwt3/Rwt4* wheat. *Lolium* isolate TP2 was avirulent on *Rwt3* cultivars but virulent on *rwt3* cultivars (Fig. 2B). When *PWT3* was disrupted (fig. S5), the resulting strain TP2 Δ 3 gained virulence on the *Rwt3* cultivars (Fig. 2B and fig. S6). *Avena* isolate Br58 was avirulent on the *Rwt3* and/or *Rwt4* carriers and virulent only on Hope (*rwt3/rwt4*) (Fig. 2C). When *PWT3* and *PWT4* were disrupted individually (figs. S5 and S7), the resulting strains Br58 Δ 3 and Br58 Δ 4 gained virulence on CS (*Rwt3/rwt4*) and Tfed (*rwt3/Rwt4*), respectively (Fig. 2C and fig. S6). Furthermore, a double disruptant gained virulence on all cultivars, including N4 (*Rwt3/Rwt4*). These results indicate that mutations or deletions of *PWT3* and *PWT4* would lead to a gain of virulence on a majority of wheat cultivars.

To determine whether mutation or loss of *PWT3* and/or *PWT4* served as key events in the evolution of wheat blast, we screened for their presence in genome sequences from a comprehensive collection of *P. oryzae* strains. *PWT4* homologs were found in some isolates but were considered non-functional (Fig. 1B and fig. S8). In contrast, *PWT3* homologs were detected in all isolates tested, with the open reading frame (ORF) sequences conforming to one of four basic types (A, B, C, and D) (Fig. 1B and fig. S9). The *Avena* and *Lolium* A-type homolog was also found in *Eleusine* isolates and some *Triticum* isolates (Fig. 1B). Upstream sequences further divided the A type into three subtypes (Ao, Ao', and Atm). Compared with the original Br58 type (Ao), the *Eleusine* isolates' type (Ao') had one base substitution in the upstream region. Three *Triticum* isolates (Fig. 1) represented three distinct types: the Ao type; the Atm type, carrying an insertion of the LINE-like element MGR583 (11) (fig. S10A); and the B type, which had 13 single-nucleotide polymorphisms (SNPs) causing 12 amino acid substitutions in ORFs (fig. S9A). Among 19 *Triticum* isolates collected in Brazil from 1990 to 1992, six, six, and seven isolates contained the Ao, Atm, and B types, respectively (table S2).

The Atm and B carriers were virulent on the four test cultivars, whereas the Ao carriers were avirulent on the *Rwt3* cultivars (table S2 and fig. S6A). An in planta expression analysis revealed that the Ao- and B-type genes in *Triticum* isolates were as highly expressed as the Ao type in Br58 (fig. S10B). In contrast, the expression level of the Atm type was reduced by a factor of 10 (fig. S10B). These results indicate that the MGR583 insertion in the Atm type has compromised its avirulence function by reducing expression.

To determine the origins of the highly divergent *PWT3* types, we screened ~100 representative *Pyricularia* isolates from various hosts and found that an isolate from *Brachiaria plantaginea* (Br35) collected in Brazil in 1990 carried the B-type *PWT3*

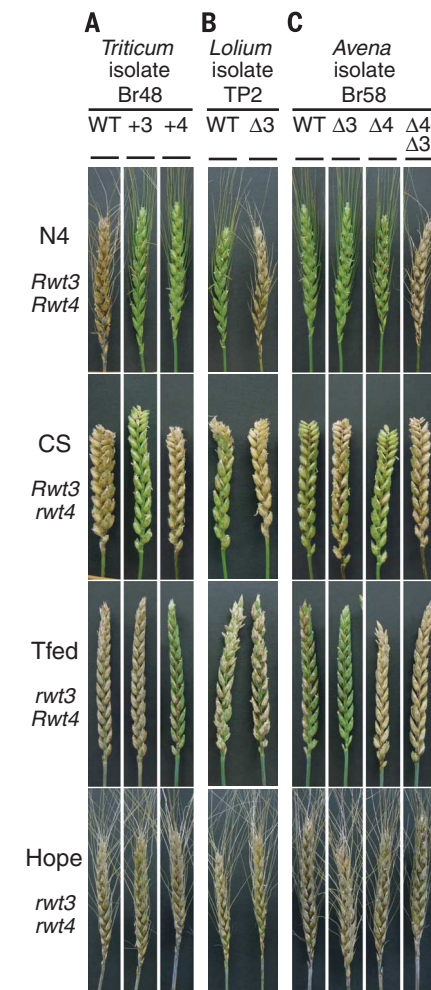


Fig. 2. *PWT3* and *PWT4* serve as the host species specificity barrier for wheat. Spikes of wheat cultivars Norin 4 (N4), Chinese Spring (CS), Transfed (Tfed), and Hope were inoculated with wild types (WT) of *Triticum* (A), *Lolium* (B), and *Avena* (C) isolates; their transformants carrying introduced *PWT3* (+3) or *PWT4* (+4); and disruptants of *PWT3* (Δ 3), *PWT4* (Δ 4), or double disruptant (Δ 3 Δ 4). Inoculated spikes were incubated for 8 days.

with 100% nucleotide sequence identity to that of Br48. However, it seemed unlikely that such a *Brachiaria* isolate was a direct ancestor of Br48 because Br35 was phylogenetically remote from Br48 (fig. S11). Comparative analyses of whole-genome sequences suggested that the direct ancestor of Br48 inherited a 1.6-Mb chromosomal segment carrying the B-type *PWT3* from a Br35-like *Brachiaria* isolate (fig. S12).

The *Triticum* isolates employed above were early isolates collected from 1990 to 1992 (Kobe University collection). To reveal population dynamics, we performed *PWT3* typing with more recent *Triticum* isolates preserved at universities or institutes in the Americas and genome sequences in public databases. Kentucky strain WBKY11-15 collected from wheat in 2011 had a previously unrecognized type (Atp) composed

of an A-type *PWT3* ORF and an inverted repeat transposon Pot3 (12) inserted into its upstream region (Fig. 3A). Bolivian strain B71 collected in 2012 had another type (Atc) in which the *PWT3*-ORF (with two nucleotide substitutions, in comparison with the A type) was disrupted by a complex insertion of transposable elements, Pyret (13) and RETRO5 (14) (Fig. 3B). All isolates characterized from Bangladesh in 2016 carried the Atc type (Fig. 3C). The Atc type was present in the 1990s as a minor population in Brazil's southern states (Fig. 3C) and became prevalent throughout Brazil in the 2000s. It spread to other countries in South America in the 2010s and was finally transmitted to Asia and caused the outbreak of wheat blast in Bangladesh.

A critical issue is why strains with the functional Ao type *PWT3* have been isolated as *Triticum* isolates. To answer this question, we surveyed the distribution of *Rwt3* and *Rwt4* in common wheat (fig. S13). In 499 local landraces collected worldwide, *Rwt3* and *Rwt4* carriers accounted for 77 and 87%, respectively (Fig. 4 and table S3). Only 6.6% of the accessions lacked both genes. Such ubiquity appears to be a key characteristic that distinguishes resistance genes conditioning the subgroup-genus specificity from those conditioning race-cultivar specificity. When ~60 improved cultivars from the Americas were tested, however, we found a noteworthy change in the early 1980s. In the late 1970s and early 1980s, the most planted cultivar in Brazil was IAC-5 (15), a carrier of *Rwt3* (table S4). Around 1980, a new cultivar (Anahuac) was introduced to Brazil and recommended to farmers (15) because it adapted very well in nonacidic soils. Anahuac was a semi-dwarf cultivar with high yield potential but a noncarrier of *Rwt3* (table S4). In 1985, a few years after the release of Anahuac, the outbreak of wheat blast occurred (4). Such wide cultivation of *rwt3* cultivar(s) in Brazil would explain why *P. oryzae* strains with the Ao type *PWT3* have been isolated from common wheat in the early period of the wheat blast outbreak in Brazil (table S2) and thereafter (Fig. 3C).

On the basis of the results described above, we present a model for the emergence of the wheat blast fungus in Brazil (fig. S14). Widespread cultivation of *rwt3* cultivars in the early 1980s allowed certain *P. oryzae* strains to colonize common wheat and increase the population despite carrying the Ao type *PWT3*. Wheat cultivars possessing *Rwt3* were still cultivated nearby and imposed selection for mutation or loss of *PWT3*. As a result, *pwt3* strains arose through independent events involving either de novo transposon insertion or gain of mutated *PWT3* from a remote strain, which would finally establish wheat pathogens as pathogenic to the entire wheat population.

This model implies that *rwt3* cultivars served as springboards for the host jump of *P. oryzae* strains (with *PWT3;pwt4*) in Brazil. Similarly, *rwt3/rwt4* cultivars may become springboards for *Avena* isolates (*PWT3;PWT4*) to jump hosts. Taken together, it is advisable to cultivate common wheat cultivars that carry both *Rwt3* and

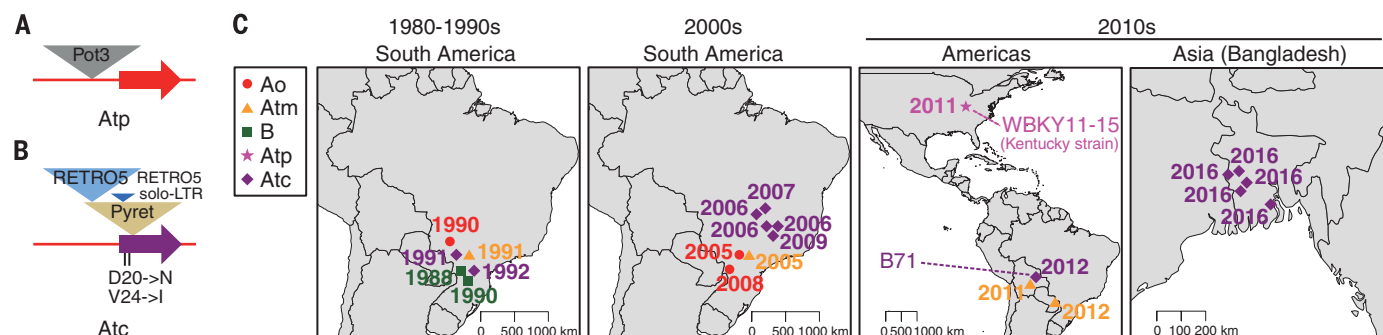


Fig. 3. Dynamics of the PWT3 types during three decades. (A) Structure of the Atp type in WBKY11-15 (LC229726), an isolate collected from wheat in Kentucky. (B) Structure of the Atc type in B71 (LC215054), a highly aggressive *Triticum* isolate collected in Bolivia. D, Asp; N, Asn; V, Val; I, Ile. (C) Geographical and historical distribution of the PWT3 types in *Triticum* isolates. Each symbol represents an isolate collected in the indicated year and is color-coded according to its PWT3 type. The Atc type appears to have expanded over time, whereas the B type was not found beyond the 1990s.

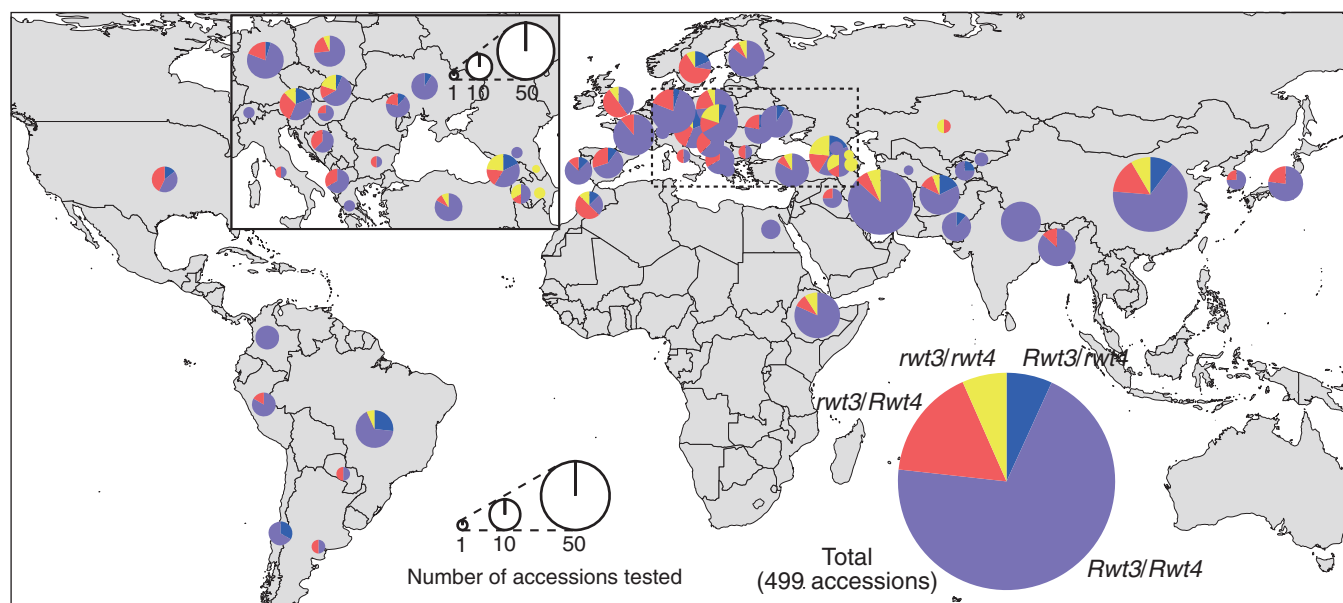


Fig. 4. Global distribution of Rwt3 and Rwt4 in local landraces of common wheat. Pie chart size is in proportion to numbers of accessions tested. The inset is a magnified map of the region inside the dashed box.

Rwt4 for forestalling reoccurrence of host jumps or at least for preventing wheat blast disease caused by infection with *Lolium* or *Avena* pathogens.

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SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/357/6346/80/suppl/DC1
Materials and Methods
Figs. S1 to S14
Tables S1 to S4
References (16–31)

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Genetic analysis of disease emergence

In the 1980s, wheat crops began to fall to the fungal pathogen that causes blast disease. First seen in Brazil, wheat blast last year caused devastating crop losses in Bangladesh. Inoue *et al.* tracked down the shifting genetics that have allowed the emergence of this potentially global threat to wheat crops (see the Perspective by Maekawa and Schulze-Lefert). Wheat varieties with a disabled resistance gene were susceptible to pathogen strains that affected oat and ryegrass crops. Subsequent genetic changes in the pathogen amped up the virulence in wheat.

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